1. The sum of the first 20 terms of the series $5+11+19+29+41+\ldots$ is
[2023 (06 Apr Shift 1)]
(1) 3520
(2) 3450
(3) 3250
(4) 3420

2. Let $a_1, a_2, a_3, \ldots, a_n$ be n positive consecutive terms of an arithmetic progression. If $d>0$ is its common difference, then $\lim_{n\to\infty}\sqrt{\frac{d}{n}}\left(\frac{1}{\sqrt{a_1}+\sqrt{a_2}}+\frac{1}{\sqrt{a_2}+\sqrt{a_3}}+\ldots+\frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}}\right)$ is
[2023 (06 Apr Shift 1)]
(1) $\frac{1}{\sqrt{d}}$
(2) $\sqrt{d}$
(3) 1
(4) 2

3. If $gcd(m,\ n)=1$ and $1^2-2^2+3^2-4^2+\ldots+(2021)^2-(2022)^2+(2023)^2=1012m^2n$ then m^2-n^2 is equal to
[2023 (06 Apr Shift 2)]
(1) 240
(2) 200
(3) 220
(4) 180

4. If $(20)^{19}+2(21)(20)^{18}+3(21)^2(20)^{17}+\ldots+20(21)^{19}=k(20)^{19}$, then k is equal to _____.
[2023 (06 Apr Shift 2)]

5. Let $S_K=\frac{1+2+\ldots+K}{K}$ and $\sum_{j=1}^{n}S^2{}_j=\frac{n}{A}\left(Bn^2+Cn+D\right)$ where $A,\ B,\ C,\ D\ \in\ N$ and A Has least value then
[2023 (08 Apr Shift 1)]
(1) $A+C+D$ is not divisible by D
(2) $A+B=5(D-C)$
(3) $A+B+C+D$ is divisible by 5
(4) $A+B$ is divisible by D

6. Let a_n be n^{th} term of the series $5+8+14+23+35+50+\ldots\ldots$and $S_n=\sum_{k=1}^{n}a_k$. Then $S_{30}-a_{40}$ is equal to
[2023 (08 Apr Shift 2)]
(1) 11310
(2) 11260
(3) 11290
(4) 11280

7. Let $0<z<y<x$ be three real numbers such that $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ are in an arithmetic progression and $x, \sqrt{2}y, z$ are in a geometric progression. If $xy+yz+zx=\frac{3}{\sqrt{2}}xyz$, then $3\left(x+y+z\right)^2$ is equal to
[2023 (08 Apr Shift 2)]

8. Let the first term a and the common ratio r of a geometric progression be positive integers. If the sum of squares of its first three terms is 33033, then the sum of these three terms is equal to
[2023 (10 Apr Shift 1)]
(1) 241
(2) 231
(3) 210
(4) 220

9. If $f(x)=\frac{(\tan 1^\circ)x+\log_e(123)}{x\ \log_e(1234)-(\tan 1^\circ)}$, $x>0$, then the least value of $f(f(x))+f\left(f\left(\frac{4}{x}\right)\right)$ is
[2023 (10 Apr Shift 1)]
(1) 0
(2) 8
(3) 2
(4) 4

10. The sum of all those terms, of the arithmetic progression $3,\ 8,\ 13,\ \ldots,\ 373$, which are not divisible by 3, is equal to ________.
[2023 (10 Apr Shift 1)]

11. If $S_n = 4 + 11 + 21 + 34 + 50 + \ldots$ to n terms, then $\frac{1}{60}(S_{29} - S_9)$ is equal to

[2023 (10 Apr Shift 2)]

(1) 223
(2) 226
(3) 220
(4) 227

12. Suppose $a_1, a_2, 2, a_3, a_4$ be in an arithmetico-geometric progression. If the common ratio of the corresponding geometric progression is 2 and the sum of all 5 terms of the arithmetico-geometric progression is $\frac{49}{2}$, then a_4 is equal to ______________

[2023 (10 Apr Shift 2)]

13. Let $x_1, x_2, \ldots, x_{100}$ be in an arithmetic progression, with $x_1 = 2$ and their mean equal to 200 . If $y_i = i(x_i - i), 1 \leq i \leq 100$, then the mean of $y_1, y_2, \ldots, y_{100}$ is

[2023 (11 Apr Shift 1)]

(1) 10100
(2) 10101.50
(3) 10049.50
(4) 10051.50

14. Let $S = 109 + \frac{108}{5} + \frac{107}{5^2} + \ldots + \frac{2}{5^{107}} + \frac{1}{5^{108}}$. Then the value of $\left(16S - (25)^{-54}\right)$ is equal to

[2023 (11 Apr Shift 1)]

15. Let a, b, c and d be positive real numbers such that $a + b + c + d = 11$. If the maximum value of $a^5 b^3 c^2 d$ is 3750β, then the value of β is

[2023 (11 Apr Shift 2)]

(1) 90
(2) 110
(3) 55
(4) 108

16. For $k \in \mathbb{N}$, if the sum of the series $1 + \frac{4}{k} + \frac{8}{k^2} + \frac{13}{k^3} + \frac{19}{k^4} + \ldots\ldots$ is 10, then the value of k is

[2023 (11 Apr Shift 2)]

17. Let $< a_n >$ be a sequence such that $a_1 + a_2 + \ldots + a_n = \frac{n^2+3n}{(n+1)(n+2)}$. If $28 \sum_{k=1}^{10} \frac{1}{a_k} = p_1 \, p_2 \, p_3 \ldots \, p_m$, where $p_1, \; p_2, \; \ldots \; p_m$ are the first m prime numbers, then m is equal to

[2023 (12 Apr Shift 1)]

(1) 5
(2) 8
(3) 6
(4) 7

18. Let $s_1, s_2, s_3, \ldots, s_{10}$ respectively be the sum of 12 terms of 10 A. Ps whose first terms are $1, \; 2, \; 3, \ldots, 10$ and the common differences are $1, \; 3, \; 5, \ldots, 19$ respectively. Then $\sum_{i=1}^{10} s_i$ is equal to

[2023 (13 Apr Shift 1)]

(1) 7220
(2) 7360
(3) 7260
(4) 7380

19. The sum to 20 terms of the series $2 \cdot 2^2 - 3^2 + 2 \cdot 4^2 - 5^2 + 2 \cdot 6^2 - \ldots\ldots\ldots\ldots$ is equal to__________.

[2023 (13 Apr Shift 1)]

20. Let $a_1, a_2, a_3, \ldots$ be a G.P. of increasing positive numbers. Let the sum of its 6^{th} and 8^{th} terms be 2 and the product of its 3^{rd} and 5^{th} terms be $\frac{1}{9}$. Then $6(a_2 + a_4)(a_4 + a_6)$ is equal to

[2023 (13 Apr Shift 2)]

(1) 3
(2) $3\sqrt{3}$
(3) 2
(4) $2\sqrt{2}$

21. Let $[\alpha]$ denote the greatest integer $\leq \alpha$. Then $\left[\sqrt{1}\right] + \left[\sqrt{2}\right] + \left[\sqrt{3}\right] + \ldots\ldots\ldots\ldots + \left[\sqrt{120}\right]$ is equal to

[2023 (13 Apr Shift 2)]

22. Let $f(x) = \sum_{k=1}^{10} k \cdot x^k, \; x \in \mathbb{R}$, if $2f(2) + f'(2) = 119(2)^n + 1$ then n is equal to ______.

[2023 (13 Apr Shift 2)]

23. Let A_1 and A_2 be two arithmetic means and $G_1,\ G_2$ and G_3 be three geometric means of two distinct positive numbers. Then $G_1^4+G_2^4+G_3^4+G_1^2G_3^2$ is equal to
[2023 (15 Apr Shift 1)]

(1) $(A_1+A_2)^2G_1G_3$
(2) $2(A_1+A_2)G_1G_3$
(3) $(A_1+A_2)G_1^2G_3^2$
(4) $2(A_1+A_2)G_1^2G_3^2$

24. If the sum of the series
$\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{2^2}-\frac{1}{2\cdot 3}+\frac{1}{3^2}\right)+\left(\frac{1}{2^3}-\frac{1}{2^2\cdot 3}+\frac{1}{2\cdot 3^2}-\frac{1}{3^3}\right)+\left(\frac{1}{2^4}-\frac{1}{2^3\cdot 3}+\frac{1}{2^2\cdot 3^2}-\frac{1}{2\cdot 3^3}+\frac{1}{3^4}\right)+\ldots\ldots$ is $\frac{\alpha}{\beta}$, where α and β are co-prime, then $\alpha+3\beta$ is equal to ______.

[2023 (15 Apr Shift 1)]

ANSWER KEYS

1. (1)	**2.** (3)	**3.** (1)	**4.** (400)	**5.** (4)	**6.** (3)	**7.** (150)	**8.** (2)
9. (3)	**10.** (9525)	**11.** (1)	**12.** (16)	**13.** (3)	**14.** (2175)	**15.** (1)	**16.** (2)
17. (3)	**18.** (3)	**19.** (1310)	**20.** (1)	**21.** (825)	**22.** (10)	**23.** (1)	**24.** (7)

1. (1)

Let the given series be S_n, then we can write as

$S_n = 5 + 11 + 19 + 29 + 41 + \ldots\ldots\ldots\ldots + T_n$

$S_n = \quad 5 + 11 + 19 + 29 + 41 + \ldots\ldots + T_{n-1} + T_n$

Subtracting above equations, we get

$0 = 5 + 6 + 8 + 10 + \ldots\ldots - T_n$

$\Rightarrow T_n = 5 + \left[\frac{(n-1)}{2}(12 + (n-2)2)\right]$

$\Rightarrow T_n = 5 + (n-1)(n+4)$

$\Rightarrow T_n = n^2 + 3n + 1$

So,

$S_{20} = \sum_{n=1}^{20} T_n$

$\Rightarrow S_{20} = \sum_{n=1}^{20}(n^2 + 3n + 1)$

$\Rightarrow S_{20} = \frac{20\times21\times41}{6} + \frac{3\times20\times21}{2} + 20$

$\Rightarrow S_{20} = 3520$

2. (3)

Given,

$a_1, a_2, a_3, \ldots, a_n$ are terms of an A. P,

So common difference will be,

$d = a_2 - a_1 = a_3 - a_2 = \ldots\ldots\ldots = a_n - a_{n-1}$

Now solving,

$\lim_{n\to\infty}\sqrt{\frac{d}{n}}\left(\frac{1}{\sqrt{a_1}+\sqrt{a_2}} + \frac{1}{\sqrt{a_2}+\sqrt{a_3}} + \ldots + \frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}}\right)$

$= \lim_{n\to\infty}\sqrt{\frac{d}{n}}\left(\frac{\sqrt{a_2}-\sqrt{a_1}}{a_2-a_1} + \frac{\sqrt{a_3}-\sqrt{a_2}}{a_3-a_2} + \ldots + \frac{\sqrt{a_n}-\sqrt{a_{n-1}}}{a_n-a_{n-1}}\right)$

$= \lim_{n\to\infty}\sqrt{\frac{d}{n}} \times \frac{1}{d}\left(\sqrt{a_n} - \sqrt{a_1}\right)$

Now using the formula $a_n = a_1 + (n-1)d$ we get,

$= \lim_{n\to\infty}\frac{1}{\sqrt{d}}\left(\frac{\sqrt{a_1+(n-1)d}-\sqrt{a_1}}{\sqrt{n}}\right)$

$= \lim_{n\to\infty}\frac{1}{\sqrt{d}}\left(\frac{\sqrt{n}\left(\sqrt{\frac{a_1}{n}+d-\frac{d}{n}}-\sqrt{\frac{a_1}{n}}\right)}{\sqrt{n}}\right)$

$= \frac{1}{\sqrt{d}}\left(\left(\sqrt{0+d-0} - \sqrt{0}\right)\right)$

$= \frac{1}{\sqrt{d}} \times \sqrt{d} = 1$

3. (1)

Let

$S = 1^2 - 2^2 + 3^2 - 4^2 + \ldots\ldots + (2021)^2 - (2022)^2 + (2023)^2$

$\Rightarrow S = (1-2)(1+2) + (3-4)(3+4) + \ldots\ldots + (2021-2022)(2021+2022) + (2023)^2$

$\Rightarrow S = -[3 + 7 + 11 + 15 + \ldots\ldots + 4043] + (2023)^2$

The number of terms in the bracket are $\frac{2022}{2} = 1011 \Rightarrow S = -\frac{1011}{2}(6 + 1010\times4) + (2023)^2$

$\Rightarrow S = -1011\times2023 + (2023)^2$

$\Rightarrow S = 2023\times1012$

$\Rightarrow S = 17^2\times7\times1012$

So,

$m = 17,\ n = 7$ and $gcd(17, 7) = 1$

Hence, $m^2 - n^2 = 17^2 - 7^2 = 240$

Hence this is the correct option.

4. (400)

Let,

$S=(20)^{19}+2(21)(20)^{18}+3(21)^{2}(20)^{17}+\ldots+20(21)^{19}$

$\Rightarrow S=20^{19}+2\cdot(20)^{19}\cdot\left(\frac{21}{20}\right)+3(20)^{19}\left(\frac{21}{20}\right)^{2}+\ldots+20(20)^{19}\cdot\left(\frac{21}{20}\right)^{19}$

$\Rightarrow S=20^{19}\left(1+2\cdot\frac{21}{20}+3\left(\frac{21}{20}\right)^{2}+\ldots+20\left(\frac{21}{20}\right)^{19}\right)$

Now on comparing with $S=k\cdot(20)^{19}$ we get,

$\Rightarrow\ k=1+2\cdot\left(\frac{21}{20}\right)+3\left(\frac{21}{20}\right)^{2}+\ldots.+20\left(\frac{21}{20}\right)^{19}\ \ldots.(1)$

$\Rightarrow\frac{21}{20}k=\frac{21}{20}+2\left(\frac{21}{20}\right)^{2}+\ldots+19\left(\frac{21}{20}\right)^{19}+20\left(\frac{21}{20}\right)^{20}\ \ \ldots.(2)$

Now on subtracting above equations we get,

$\Rightarrow k-\frac{21}{20}k=1+\frac{21}{20}+\left(\frac{21}{20}\right)^{2}+\ldots+\left(\frac{21}{20}\right)^{19}-20\left(\frac{21}{20}\right)^{20}$

$\Rightarrow\frac{-k}{20}=\frac{\left(\frac{21}{20}\right)^{20}-1}{\frac{21}{20}-1}-20\left(\frac{21}{20}\right)^{20}$

$\Rightarrow-\frac{k}{20}=\left(\left(\frac{21}{20}\right)^{20}-1\right)\times20-20\left(\frac{21}{20}\right)^{20}$

$\Rightarrow\frac{-k}{20}=-20$

$\Rightarrow k=400$

5. (4)

Given that $S_K=\frac{1+2+\ldots+K}{K}$

We know that $1+2+3.\ldots+n=\frac{n(n+1)}{2}$

$\Rightarrow S_K=\frac{K(K+1)}{2K}=\frac{K+1}{2}$

Now,

$\sum_{j=1}^{n}\left(S_j\right)^{2}=\sum_{j=1}^{n}\frac{1}{4}\left(2^{2}+3^{2}+4^{2}\ldots(n+1)^{2}\right)$

$=\sum_{j=1}^{n}\frac{1}{4}\left(1^{2}+2^{2}+3^{2}+4^{2}\ldots(n+1)^{2}-1^{2}\right)$

$\Rightarrow\frac{1}{4}\left(\frac{(n+1)(n+2)(2n+3)}{6}-1\right)$

$=\frac{1}{4}\left(\frac{(n^{2}+3n+2)(2n+3)-6}{6}\right)$

$=\frac{1}{4}\left(\frac{2n^{3}+6n^{2}+4n+3n^{2}+9n+6-6}{6}\right)$

$=\frac{1}{4}\left(\frac{2n^{3}+9n^{2}+13n}{6}\right)$

$=\frac{n}{24}\left(2n^{2}+9n+13\right)$

On comparing this with $\sum_{j=1}^{n}S^{2}{}_{j}=\frac{n}{A}\left(Bn^{2}+Cn+D\right)$ we get,

$A=24,\ B=2,\ C=9,\ D=13$

$\Rightarrow\frac{A+B}{D}=\frac{26}{13}=2$

That means, $A+B$ is divisible by D.

Hence this is the correct option.

6. (3)

The given series is a_n, then we can write as
$a_n = 5 + 8 + 14 + 23 + \ldots\ldots\ldots\ldots + a_n$
$a_n = \quad 5 + 8 + 14 + 23 + \ldots\ldots + a_{n-1} + a_n$
Subtracting above equations, we get
$0 = 5 + 3 + 6 + 9 + \ldots. + (a_n - a_{n-1}) - a_n$
$\Rightarrow a_n = 5 + \left[\frac{(n-1)}{2}(2 \times 3 + (n-1-1)3)\right]$
$\Rightarrow a_n = 5 + \frac{(n-1)}{2}(3n)$
$\Rightarrow a_n = \frac{1}{2}(3n^2 - 3n + 10)$
So,
$\Rightarrow a_{40} = \frac{1}{2}(3 \times 40^2 - 3 \times 40 + 10)$
$\Rightarrow a_{40} = \frac{1}{2}(4800 - 120 + 10)$
$\Rightarrow a_{40} = 2345$
$\Rightarrow S_{30} = \sum_{n=1}^{30} a_n$
$\Rightarrow S_{30} = \frac{1}{2}\left(3\sum_{n=1}^{30}(n^2) - 3\sum_{n=1}^{30}(n) + \sum_{n=1}^{30} 10\right)$
$= \frac{1}{2}\left(\frac{3(30)(31)(61)}{6} - \frac{3(30)(31)}{2} + 10 \times 30\right)$
$\Rightarrow S_{30} = 13635$
$\Rightarrow S_{30} - a_{40} = 13635 - 2345 = 11290$
Therefore, the required value is 11290.

7. (150)

Given that $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ are in AP and $x, \sqrt{2}y, z$ are in GP.
As given, $\frac{2}{y} = \frac{1}{x} + \frac{1}{y}$ ....(i)
Also, $2y^2 = xz$ ....(ii)
Also given that $xy + yz + zx = \frac{3}{\sqrt{2}}xyz$
$\Rightarrow \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{3}{\sqrt{2}}$ .....(iii)
From (i) and (iii) we get $\frac{3}{y} = \frac{3}{\sqrt{2}}$
$y = \sqrt{2}$ .....(iv)
Now from (ii) $xz = 4$ ....(v)
Now using (ii), (iv) and (v)
$\Rightarrow x + z = 4\sqrt{2}$
Hence $3\left(x + y + z\right)^2 = 3\left(\sqrt{2} + 4\sqrt{2}\right)^2$
$= 150$
Therefore, this is the required answer.

8. (2)

Given that,
$a^2 + (ar)^2 + \left(ar^2\right)^2 = 33033,\ (a, r \in N)$
$\Rightarrow a^2\left(1 + r^2 + r^4\right) = 11^2 \times 273$
On comparing both sides of the equation we get,
$a = 11$ and $1 + r^2 + r^4 = 273$
$\Rightarrow r^2 + r^4 = 272$
$\Rightarrow r^2\left(1 + r^2\right) = 16 \times 17$
$\Rightarrow r = 4$
Now $a + ar + ar^2 = 11 + 11 \times 4 + 11 \times 16$
$= 11 + 44 + 176 = 231.$
Therefore, the required value is 231

9. (3)

Given,

$f(x)=\frac{(\tan 1^{\circ})x+\log_e(123)}{x\log_e(1234)-(\tan 1^{\circ})}, x>0$

Now, let $\tan 1^{\circ}=a, \log_e(123)=b$ and $\log_e(1234)=c$

So, $f(x)=\frac{ax+b}{cx-a}$

$\Rightarrow f(f(x))=\frac{af(x)+b}{cf(x)-a}$

$\Rightarrow f(f(x))=\frac{a\left(\frac{ax+b}{cx-a}\right)+b}{c\left(\frac{ax+b}{cx-a}\right)-a}$

$\Rightarrow f(f(x))=\frac{a^2x+ab+b(cx-a)}{acx+bc-a(cx-a)}=\frac{a^2x+bcx}{bc+a^2}=x$

$\Rightarrow f(f(x))+f\left(f\left(\frac{4}{x}\right)\right)=x+\frac{4}{x}$

$\because x>0$, then least value $=\sqrt{x\cdot\frac{4}{x}}=2$

10. (9525)

Given sequence is $3, 8, 13, \ldots\ldots 373$.

Here $a=3, d=5$ and $a_n=373$

Now $a_n=a+(n-1)d$

$\Rightarrow 373=3+(n-1)5$

$\Rightarrow n=75$

We know that $S_n=\frac{n}{2}(a+a_n)$

$\Rightarrow S_{75}=\frac{75}{2}(3+373)$

$\Rightarrow S_{75}=14100$

Now let us write the sequence of terms which are divisible by 3.

We get, $3, 18, 33, \ldots\ldots 363$

$\Rightarrow 363=3+(n-1)15$

$\Rightarrow n=25$

Now let us find the sum of terms divisible by 3.

$\Rightarrow S_{div\ by\ 3}=\frac{25}{2}(3+363)=4575$

Required sum $=$Sum of 75 terms$-$Sum of terms divisible by 3

$=14100-4575$

$=9525$.

Therefore, the required sum is 9525

11. (1)

Given,

$S_n = 4 + 11 + 21 + 34 + \ldots\ldots\ldots\ldots + T_n$

$S_n = \quad 4 + 11 + 21 + 34 \ldots\ldots + T_{n-1} + T_n$

Subtracting above equations, we get

$0 = 4 + 7 + 10 + 13 + \ldots . - T_n$

$\Rightarrow T_n = 4 + 7 + 10 + 13 + \ldots$

The above series is in AP.

$\Rightarrow T_n = \frac{n}{2}(2 \times 4 + (n-1)3)$

$\Rightarrow T_n = \frac{n}{2}(3n + 5)$

So,

$S_n = \sum \frac{3n^2+5n}{2}$

$\Rightarrow S_n = \frac{1}{2}\left(\frac{3n(n+1)(2n+1)}{6} + \frac{5n(n+1)}{2}\right)$

$\Rightarrow S_n = \frac{n(n+1)}{2}\left(\frac{(2n+1)}{2} + \frac{5}{2}\right)$

$\Rightarrow S_{29} = \frac{29 \times 30}{2}\left(\frac{59}{2} + \frac{5}{2}\right)$

$\Rightarrow S_{29} = 29 \times 15 \times 32 = 13920$

$\Rightarrow S_9 = \frac{9 \times 10}{2}\left(\frac{19}{2} + \frac{5}{2}\right)$

$\Rightarrow S_9 = 9 \times 5 \times 12 = 540$

$\Rightarrow \frac{S_{29} - S_9}{60} = \frac{13920 - 540}{60} = 223.$

Therefore, the required answer is 223.

12. (16)

Given,

$a_1, a_2, 2, a_3, a_4$ be in an arithmetico-geometric progression,

And the common ratio of the corresponding geometric progression is 2 and the sum of all 5 terms of the arithmetico-geometric progression is $\frac{49}{2}$

So,common ratio Here $r = 2$ and let common difference $= d$

And we know that,

$a_2 = (a_1 + d) \times 2$

$2 = (a_1 + 2d) \times 2^2 \ldots\ldots\ldots(i)$

$a_3 = (a_1 + 3d) \times 2^3$

$a_4 = (a_1 + 4d) \times 2^4$

Now adding all above equations we get,

$31a_1 + 98d = \frac{49}{2} \ldots\ldots\ldots(ii)$

Now from (i) & (ii)

$a_1 = 0, d = \frac{1}{4}$

Hence, $a_4 = 16(0 + 1) = 16$

13. (3)

Given,

Mean of $x_1,\ x_2 \ldots\ldots\ldots\ldots, x_{100}$ is 200

So, the sum of observation will be,

$\sum x_i = 100 \times 200$

Now using the sum of $A.P$ formula in above equation as all terms are in arithmetic progression we get,

$\frac{100}{2}(x_1 + x_{100}) = 100 \times 200$

$\Rightarrow 50(2 + x_{100}) = 100 \times 200$

$\Rightarrow x_{100} = 398$

$\Rightarrow x_1 + 99d = 398$

$\Rightarrow d = 4$

Now, $x_i = 2 + (i-1)4 = 4i - 2$

So, $y_i = i(x_i - i) = 3i^2 - 2i$

Now finding mean we get,

$\bar{y} = \frac{1}{100}\sum y_i$

$\Rightarrow \bar{y} = \frac{1}{100}\sum(3i^2 - 2i)$

$\Rightarrow \bar{y} = \frac{1}{100}\left[3 \times \frac{100 \times 101 \times 201}{6} - 2\frac{100 \times 101}{2}\right]$

$\Rightarrow \bar{y} = \left[\frac{101 \times 201}{2} - 101\right]$

$\Rightarrow \bar{y} = 10049.5$

14. (2175)

Given,

$S = 109 + \frac{108}{5} + \frac{107}{5^2} + \ldots + \frac{2}{5^{107}} + \frac{1}{5^{108}} \ldots.(1)$

$\frac{S}{5} = \frac{109}{5} + \frac{108}{5^2} + \ldots + \ldots + \frac{2}{5^{108}} + \frac{1}{5^{109}} \ldots.(2)$

Now subtracting above equations we get,

$\frac{-4S}{5} = -109 + \left(\frac{1}{5} + \frac{1}{5^2} + \ldots + \frac{1}{5^{108}} + \frac{1}{5^{109}}\right)$

$\Rightarrow \frac{-4S}{5} = -109 + \frac{1}{5}\frac{\left(1 - \left(\frac{1}{5}\right)^{109}\right)}{1 - \frac{1}{5}}$

$\Rightarrow \frac{4S}{5} = -109 + \frac{1}{4}\left(1 - \frac{1}{5^{109}}\right)$

$\Rightarrow \frac{16S}{5} = -436 + \left(1 - \frac{1}{5^{109}}\right)$

$\Rightarrow 16S = -436 \times 5 + \left(5 - 5^{-108}\right)$

$\Rightarrow 16S - (25)^{-54} = 2175$

15. (1)

Given:

$a + b + c + d = 11$

We know that,

$\mathrm{AM} \geq \mathrm{GM}$

$\Rightarrow \frac{\frac{a}{5} + \frac{a}{5} + \frac{a}{5} + \frac{a}{5} + \frac{a}{5} + \frac{b}{3} + \frac{b}{3} + \frac{b}{3} + \frac{c}{2} + \frac{c}{2} + \frac{d}{1}}{11} \geq \sqrt[11]{\frac{a^5 b^3 c^2 d}{5^5 3^3 2^2}}$

$\Rightarrow \frac{a+b+c+d}{11} \geq \sqrt[11]{\frac{a^5 b^3 c^2 d}{5^5 3^3 2^2}}$

$\Rightarrow \frac{11}{11} \geq \sqrt[11]{\frac{a^5 b^3 c^2 d}{5^5 3^3 2^2}}$

$\Rightarrow \sqrt[11]{\frac{a^5 b^3 c^2 d}{5^5 3^3 2^2}} \leq 1$

$\Rightarrow a^5 b^3 c^2 d \leq 5^5 3^3 2^2$

$\Rightarrow a^5 b^3 c^2 d \leq 90 \cdot 3750$

So,

$\beta = 90$

16. (2)

Given,

$10=1+\frac{4}{k}+\frac{8}{k^2}+\frac{13}{k^3}+\frac{19}{k^4}\ldots\ldots\ldots\infty\ldots\ldots(1)$

$\frac{10}{k}=\frac{1}{k}+\frac{4}{k^2}+\frac{8}{k^3}+\frac{13}{k4}+\ldots\ldots\ldots\infty\ldots\ldots(2)$

On subtracting above two equations we get,

$10-\frac{10}{k}=1+\frac{3}{k}+\frac{4}{k^2}+\frac{5}{k^3}+\frac{6}{k^4}\ldots\ldots\ldots\infty$

$\Rightarrow 10\left(1-\frac{1}{k}\right)=1+\frac{3}{k}+\frac{4}{k^2}+\frac{5}{k^3}+\frac{6}{k^4}\ldots\ldots\ldots\infty\ldots\ldots(3)$

$\Rightarrow 10\left(1-\frac{1}{k}\right)\left(\frac{1}{k}\right)=\ \frac{1}{k}+\frac{3}{k^2}+\frac{4}{k^3}+\frac{5}{k^4}+\frac{6}{k^5}\ldots\ldots\ldots\infty\ldots\ldots(4)$

Again subtracting both equations we get,

$\Rightarrow 10\left(1-\frac{1}{k}\right)^2=1+\frac{2}{k}+\frac{1}{k^2}+\frac{1}{k^3}+\frac{1}{k^4}\ldots\ldots\ldots\infty$

$\Rightarrow 10\left(1-\frac{1}{k}\right)^2=\left(1+\frac{1}{k}+\frac{1}{k^2}+\frac{1}{k^3}+\frac{1}{k^4}\ldots\ldots\ldots\infty\right)+\frac{1}{k}$

$\Rightarrow 10\left(1-\frac{1}{k}\right)^2=\left(\frac{1}{1-\frac{1}{k}}\right)+\frac{1}{k}$

$\Rightarrow 10(k-1)^3=k\left(k^2+k-1\right)$

$\Rightarrow 9k^3-31k^2+31k-10=0$

$\Rightarrow k=2$

Hence, $k=2$

17. (3)

Given,

$S_n=\sum_{i=1}^{n}a_i=\frac{n^2+3n}{(n+1)(n+2)}$

Now we know that,

$a_n=S_n-S_{n-1}$

$\Rightarrow a_n=\frac{n^2+3n}{(n+1)(n+2)}-\frac{(n-1)^2+3(n-1)}{n(n+1)}$

$\Rightarrow a_n=\frac{4}{n(n+1)(n+2)}$

Now solving, $\sum_{k=1}^{10}\frac{1}{a_k}=\frac{1}{4}\sum_{k=1}^{10}k(k+1)(k+2)$

$\Rightarrow \sum_{k=1}^{10}\frac{1}{a_k}=\frac{1}{16}\sum_{k=1}^{10}k(k+1)(k+2)(k+3)-(k-1)k(k+1)(k+2)$

$\Rightarrow \sum_{k=1}^{10}\frac{1}{a_k}=\frac{1}{16}[(1\cdot2\cdot3\cdot4.-0)+(2\cdot3\cdot4\cdot5-1\cdot2\cdot3\cdot4)\ldots.+(10\cdot11\cdot12\cdot13-9\cdot10\cdot11\cdot12)]$

$\Rightarrow \sum_{k=1}^{10}\frac{1}{a_k}=\frac{1}{16}[10\cdot11\cdot12\cdot13-0]$

$\Rightarrow \sum_{k=1}^{10}\frac{1}{a_k}=\frac{1}{2}(5\cdot11\cdot3\cdot13)$

$\Rightarrow\ 28\sum_{k=1}^{10}\frac{1}{a_k}=\frac{28\times5\times11\times3\times13}{2}$

$\Rightarrow 28\sum_{k=1}^{10}\frac{1}{a_k}=2\cdot3\cdot5\cdot7\cdot11\cdot13$

Hence, there are six prime numbers in multiplications,

So, $m=6$

18. (3)

Given,

First term of A.P are $1,2,3.\ldots.10$, so general term of the first term will be i

And common difference are $1,3,5,\ldots.$, so general term of common difference is given by $2i-1$

Now sum of the A.P is given by,

$S_i=\frac{12}{2}[2\times i+(12-1)(2i-1)]$

$\Rightarrow S_i=6[2\times i+11(2i-1)]$

$\Rightarrow S_i=144i-66$

So, $\sum_{i=1}^{10}S_i=\sum_{i=1}^{10}144i-66\sum_{i=1}^{10}1$

$\Rightarrow \sum_{i=1}^{10}S_i=144\left(\frac{10\times11}{2}\right)-66\times10$

$\Rightarrow \sum_{i=1}^{10}S_i=792(10)-66\times10$

$\Rightarrow \sum_{i=1}^{10}S_i=7260$

Hence this is the correct option.

19. (1310)

Given,

$2 \cdot 2^2 - 3^2 + 2 \cdot 4^2 - 5^2 + 2 \cdot 6^2 - \ldots\ldots\ldots\ldots 20$ terms,

$= \sum_{r=1}^{10}\left(2 \cdot (2r)^2 - (2r+1)^2\right)$

$= \sum_{r=1}^{10}\left(8r^2 - 4r^2 - 4r - 1\right)$

$= \sum_{r=1}^{10}\left(4r^2 - 4r - 1\right)$

$= 4\sum_{r=1}^{10} r^2 - 4\sum_{r=1}^{10} r - \sum_{r=1}^{10} 1$

$= \frac{4 \cdot 10 \cdot 11 \cdot 21}{6} - 4\frac{10 \cdot 11}{2} - 10$

$= 44 \cdot 35 - 220 - 10 = 1540 - 230 = 1310$

20. (1)

Given,

$a_1, a_2, a_3, \ldots.$ be a G.P. of increasing positive numbers,

And the sum of its 6^{th} and 8^{th} terms be 2

So, $a_6 + a_8 = 2$

$\Rightarrow ar^5 + ar^7 = 2 \ \ldots..(1)$

And the product of its 3^{rd} and 5^{th} terms be $\frac{1}{9}$,

So, $a_3 \cdot a_5 = \frac{1}{9} \Rightarrow a^2 \cdot r^2 \cdot r^4 = \frac{1}{9}$

$\Rightarrow ar^3 = \frac{1}{3}$

Now from putting the value of $ar^3 = \frac{1}{3}$ in equation (1) we get,

$\frac{r^2}{3} + \frac{r^4}{3} = 2$

$\Rightarrow r^4 + r^2 = 6$

$\Rightarrow \left(r^2+3\right)\left(r^2-2\right) = 0$

$\Rightarrow r^2 = 2$

$ar^3 = \frac{1}{3} \Rightarrow ar \cdot 2 = \frac{1}{3} \Rightarrow ar = \frac{1}{6}$

Now finding the value of $6(a_2 + a_4)(a_4 + a_6)$, we get

$= 6\left(ar + ar^3\right)\left(ar^3 + ar^5\right)$

$= 6\left(\frac{1}{6} + \frac{1}{3}\right)\left(\frac{1}{3} + \frac{2}{3}\right)$

$= 6 \cdot \frac{1}{2} \cdot 1 = 3$

21. (825)

Given,

$\left[\sqrt{1}\right] + \left[\sqrt{2}\right] + \left[\sqrt{3}\right] + \ldots\ldots\ldots\ldots + \left[\sqrt{120}\right]$

Now we now that,

$\left[\sqrt{x}\right] = 1,$ when $x \in [1, 4),$ $\left[\sqrt{x}\right] = 2,$ when $x \in [4, 9)$ and similarly $\left[\sqrt{x}\right] = 10,$ when $x \in [100, 120)$

Now using the above formula we get,

$E = 1 + 1 + 1 + 2 + 2 + 2 + 2 + 2 + 3 + 3 + 3 + 3 + 3$
$+ 3 + 3 + 4 + 4 + \ldots$

$\Rightarrow E = 3 \times 1 + 5 \times 2 + 7 \times 3 + \ldots.. + 19 \times 9 + 10 \times 21$

$\Rightarrow E = \sum_{r=1}^{10}(2r+1)r$

$\Rightarrow E = 2\sum_{r=1}^{10}\left(r^2 + r\right)$

$\Rightarrow E = 2\left[\frac{10 \times 11 \times 21}{6}\right] + \frac{10 \times 11}{2}$

$\Rightarrow E = 770 + 55$

$\Rightarrow E = 825$

22. (10)

Given,

$f(x)=\sum_{k=1}^{10} k\cdot x^k,\ x\in\mathbb{R}$

$\Rightarrow f(x)=x+2x^2+3x^3+\ldots\ldots\ldots+10x^{10}$

Now let $S=x+2x^2+3x^3+\ldots\ldots\ldots+10x^{10}$

So, $S\cdot x\ =\ \ x^2+2x^3+\ldots\ldots\ldots\ldots+9x^{10}+10x^{11}$

Now subtracting above equations we get,

$S(1-x)=x+x^2+x^3+\ldots\ldots+x^{10}-10x^{11}$

$\Rightarrow S(1-x)=\frac{x\left(1-x^{10}\right)}{1-x}-10x^{11}$

$\Rightarrow S=\frac{x\left(1-x^{10}\right)}{(1-x)^2}-\frac{10x^{11}}{(1-x)}=f(x)$

Now finding, $f(2)=2\left(1-2^{10}\right)+10\cdot 2^{11}$

$\Rightarrow f(2)=2+18\cdot 2^{10}$

Now finding $f'(x)$ we get,

$f'(x)=\frac{-10x^{11}}{(1-x)^2}-\frac{110x^{10}}{1-x}-\frac{10x^{10}}{(1-x)^2}+\frac{2x\left(1-x^{10}\right)}{(1-x)^3}+\frac{1-x^{10}}{(1-x)^2}$

$\Rightarrow f'(2)=\frac{-10\cdot 2^{11}}{(1-2)^2}-\frac{110\cdot 2^{10}}{1-2}-\frac{10\cdot 2^{10}}{(1-2)^2}+\frac{2\cdot 2\cdot\left(1-2^{10}\right)}{(1-2)^3}+\frac{1-2^{10}}{(1-2)^2}$

$\Rightarrow f'(2)=-10\cdot 2^{11}+110\cdot 2^{10}-10\cdot 2^{10}-4\left(1-2^{10}\right)+1-2^{10}$

$\Rightarrow f'(2)=83\cdot 2^{10}-3$

$2f(2)+f'(2)=2\left(2+18\cdot 2^{10}\right)+83\cdot 2^{10}-3=119(2)^{10}+1$

$\therefore\ \ n=10$

23. (1)

Let the two numbers are $a,\ b$

$\Rightarrow a, A_1, A_2, b$ are in AP.

$\Rightarrow b=a+(4-1)d$

$\Rightarrow d=\frac{b-a}{3}$

$\Rightarrow A_1=a+\frac{b-a}{3}=\frac{2a+b}{3}$

$\Rightarrow A_2=a+\frac{b-a}{3}\cdot 2=\frac{a+2b}{3}$

Similarly a, G_1, G_2, G_3, b are in GP.

$\Rightarrow b=a(r)^{5-1}$

$\Rightarrow r=\left(\frac{b}{a}\right)^{\frac{1}{4}}$

$\Rightarrow G_1=a\left(\frac{b}{a}\right)^{\frac{1}{4}}$

$\Rightarrow G_2=a\left(\frac{b}{a}\right)^{\frac{2}{4}}$

$\Rightarrow G_3=a\left(\frac{b}{a}\right)^{\frac{3}{4}}$

$=(G_1)^4+(G_2)^4+(G_3)^4+(G_1)^2\cdot(G_3)^2$

$\Rightarrow a^4\cdot\frac{b}{a}+a^4\cdot\frac{b^2}{a^2}+a^4\cdot\frac{b^3}{a^3}+a^4\cdot\frac{b^2}{a^2}$

$=ba^3+b^2a^2+b^3a+a^2b^2$

$=ab\left(a^2+b^2+2ab\right)=ab(a+b)^2$

$\Rightarrow (A_1+A_2)^2\cdot G_1G_3=(a+b)^2\cdot ab$

Hence this is the correct option.

24. (7)

Given,

Series $\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{2^2}-\frac{1}{2\cdot3}+\frac{1}{3^2}\right)+\left(\frac{1}{2^3}-\frac{1}{2^2\cdot3}+\frac{1}{2\cdot3^2}-\frac{1}{3^3}\right)+\left(\frac{1}{2^4}-\frac{1}{2^3\cdot3}+\frac{1}{2^2\cdot3^2}-\frac{1}{2\cdot3^3}+\frac{1}{3^4}\right)+\ldots\ldots$

Now let $a=\frac{1}{2},\ b=\frac{1}{3}$

So, series will be:

$(a-b)+\left(a^2-ab+b^2\right)+\left(a^3-a^2b+ab^2-b^3\right)+\ldots..$

$=\frac{1}{a+b}\left(\left(a^2-b^2\right)+\left(a^3+b^3\right)+\left(a^4-b^4\right)+\ldots\right)$

$=\left(\frac{1}{a+b}\right)\left(a^2+a^3+a^4\ldots-\left(b^2-b^3+b^4\ldots.\right)\right)$

Now using infinite G.P formula we get,

$=\left(\frac{1}{a+b}\right)\left(a^2+a^3+a^4\ldots-\left(b^2-b^3+b^4\ldots.\right)\right)$

$=\left(\frac{1}{a+b}\right)\left(\frac{a^2}{1-a}-\frac{b^2}{1+b}\right)$

Now putting the value of a & b we get,

$=\frac{1}{\frac{1}{2}+\frac{1}{3}}\left(\frac{\frac{1}{4}}{1-\frac{1}{2}}-\frac{\frac{1}{9}}{1+\frac{1}{3}}\right)$

$=\frac{6}{5}\left(\frac{1}{2}-\frac{1}{12}\right)$

$=\frac{6}{5}\left(\frac{5}{12}\right)=\frac{1}{2}=\frac{\alpha}{\beta}$

Hence, $\alpha+3\beta=1+6=7$